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Distribution of trap energy level in AlGaN/GaN high-electron-mobility transistors on Si under ON-state stress

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The distribution of trap energy (DTE) levels was observed in the energy band gap of buffer GaN by temperature-dependent current transient measurements on AlGaN/GaN HEMTs under fully ON drain-stress ($V_{D[ON]\_Stress}$) conditions. The activation energies (E_a's) obtained from current transients increase with increasing $V_{D[ON]\_Stress}$. Using a multitrap energy (MTE) model, the applied-$V_{D[ON]\_Stress}$-dependent E_a is attributed to DTE levels in the GaN energy band gap, rather than to discrete single trap energy levels. An effective activation energy (E_{a_eff}) corresponding to trap energy levels activated by the applied $V_{D[ON]\_Stress}$ is thus obtained. This observation is validated with two-dimensional numerical simulations. This study will help device designers develop a "DTE-dependent" emission time constant model that is readily applicable for the reliability modelling of future GaN-based circuits.

Despite the fact that AlGaN/GaN high-electron-mobility transistors (HEMTs) have shown excellent high-voltage[1)] and high-frequency[2)] performance characteristics, various underlying material-based anomalies still persist.[3,4)] The unintentionally doped GaN buffer in AlGaN/GaN HEMTs contains various defects or trap states[3–5)] that cause current collapse and other trapping-related reliability issues under device operation. Numerous techniques are being employed to determine the origin and role of the defects that affect the reliability of GaN devices. For highly scaled devices, the choice of the analysis method also becomes very critical. The multiexponential current transient technique is reported to be one of the most suitable methods for studying the trap properties of GaN transistors.[6–9)] Most of the studies based on this technique use fully OFF, semi-ON, or fully ON-state bias conditions to investigate the trapping mechanism. Fully OFF-state stress conditions facilitate the study of surface/barrier-related trapping that results in dynamic $R_{DS[ON]}$ degradation and threshold voltage shift.[9,10)] A semi-ON state bias condition resulting in the partial depletion of the channel has been adopted to study the buffer-related traps.[9)] However, the negative bias to the gate electrode will also induce some surface- and barrier-related trapping, resulting in both surface/barrier- and buffer-related components. On the other hand, under fully ON-state stress, when the drain voltage is significantly high, the channel electrons can spill over and gain sufficient energy at the gate edge of the drain side to reach deep into the GaN buffer.[11)] The energetic electrons that spread[12)] into the GaN buffer are trapped by various traps present in the GaN buffer.[13,14)] Hu and Hashizume[15)] reported that even though both surface leakage currents and hot electrons are responsible for the nonlocalized trapping effects, the trapping of hot electrons in the GaN buffer has a dominant effect under fully ON-state stress conditions. This is even more pronounced if the surface-related trapping component is suppressed through appropriate surface treatments.[2,15)] Even though long hours of ON-state stress resulted in the generation of interface- and surface-related traps due to hot electron effects,[16,17)] the physical origins of the traps have not been distinctly established. Moreover, del Alamo and Joh[18)] have reported the absence of a hot-electron-based degradation signature during their long-time stress tests and attributed "inverse piezoelectric effects" as a possible degradation mode. However, under a short-period ON-state stress condition, Hu et al.[11)] have reported a hot-electron buffer trapping effect. Thus, a short-period ON-state stress condition is more relevant for the study of buffer-related traps in AlGaN/GaN HEMTs.[4,19)]

By analyzing the revival of collapsed current using incident light with varying wavelength, Binari et al.[13)] reported on the probability of the presence of distribution of trap energy (DTE) levels. A current collapse model incorporating two distinct trap levels of 1.80 and 2.85 eV based on photoionization spectra was proposed by Klein et al.[19)] They reported that the photoconductivity spectra of the traps overlap under certain wavelengths used for the photoinduced recovery of the collapsed drain current (I_D). However, such photoionization spectral studies are more effective with gateless or semitransparent gate device structures. A thorough understanding of the DTE levels in the GaN energy band gap is still missing for gated devices. In this work, we studied the trapping dynamics under actual device operation conditions involved in AlGaN/GaN HEMTs by a temperature-dependent I_D transient measurement technique under different $V_{D[ON]\_Stress}$ conditions. We obtained different activation energies (E_a's) under different $V_{D[ON]\_Stress}$ conditions. The behavior of the time constants under different $V_{D[ON]\_Stress}$ conditions at different temperatures will be discussed. In addition, we propose that the obtained effective activation energies (E_{a_eff}'s) correspond to different energy levels in the continuous DTE function activated under different $V_{D[ON]\_Stress}$ conditions. This is further verified through TCAD[20)] two-dimensional (2D) numerical transient simulations by incorporating the experimentally determined trap parameters.

The device structures were grown on 4-in. HR-Si(111) substrates by metal organic chemical vapor deposition. The device heterostructure consists of GaN (2 nm) cap layer/$Al_{0.26}GaN$ (18 nm) barrier layer/i-GaN (800 nm)/buffer GaN (1400 nm)/AlN (~100 nm) nucleation layer/Si(111). The cap, barrier, and i-GaN layers are unintentionally doped. After mesa isolation by BCl_3/Cl_2 plasma, a nongold metal stack (Ta/Si/Ti/Al/Ni/Ta) was deposited and annealed at 800 °C for 30 s, which exhibited a contact resistance of 0.24 Ω-mm. Consecutively, the samples went through ammonium sulphide [$(NH_4)_2S_x$] surface treatment[2,21)] and 120-nm-thick SiN deposition by plasma-enhanced chemical vapor deposition (PECVD) at 300 °C. A Schottky gate was formed with Ni/Al/Ta after etching SiN. Then, the devices were

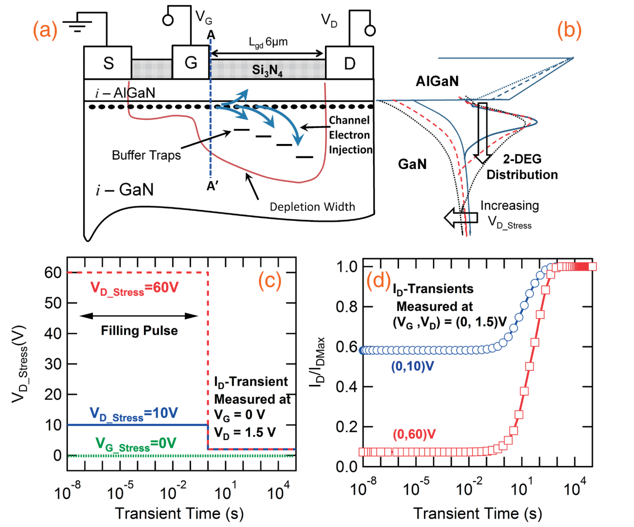
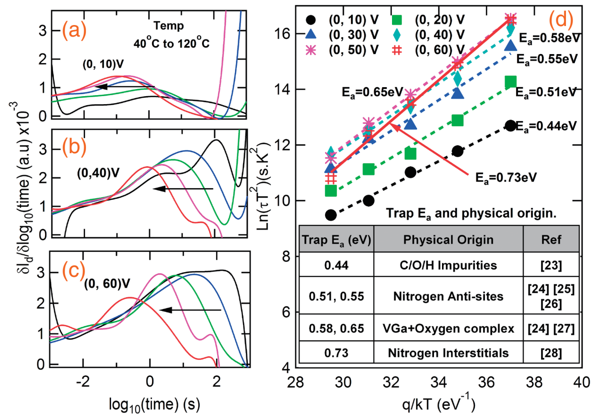


Fig. 1. (a) Schematic of AlGaN/GaN HEMT structure showing trapping of 2DEG channel electrons at high $V_{D[ON]\_Stress}$ of (0, 60) V, (b) energy band diagram of an AlGaN/GaN heterojunction showing the 2DEG distribution and band bending under different $V_{D[ON]\_Stress}$ conditions at the cut-line position of AA′, (c) time domain representation of $V_{D[ON]\_Stress}$ for 10 and 60 V, and (d) corresponding evolution of collapsed current I_D transients as a function of transient time.

passivated with 120-nm-thick PECVD SiN at 300 °C. The device dimensions were source–gate distance (L_{sg}) = 2 μm, gate length (L_g) = 2 μm, gate width (W_g) = 80 μm, and gate–drain distance (L_{gd}) = 6 μm. The devices exhibited a maximum drain current density of 540 mA/mm, a maximum extrinsic output transconductance of 187 mS/mm, and a threshold voltage of −2.5 V. The OFF-state breakdown voltage of the devices was ≈640 V. The detailed device fabrication and dynamic-$R_{DS[ON]}$ characteristics have been reported elsewhere.[22] The temperature-dependent I_D transient measurements (from 40 to 120 °C, increments of 20 °C) under different ON-state V_{D_Stress} conditions were performed using an Agilent B1505A power device analyzer. Figure 1(a) shows the schematic of the AlGaN/GaN HEMT structure and the trapping of 2D electron gas (2DEG) channel electrons under fully $V_{D[ON]\_Stress}$ conditions. The corresponding energy band diagram (cut-line position of AA′), depicting the downward band bending as well as the spread of the 2DEG wave function into the GaN buffer region, is shown in Fig. 1(b). For the measurements, the source terminal was grounded, the gate terminal was biased at ($V_{G\text{-}Stress}$) = 0 V, and the drain terminal was stressed at different voltages ($V_{D[ON]\_Stress}$) = 10 to 60 V with steps of 10 V. The $V_{D[ON]\_Stress}$ or filling pulse was maintained for a period of 1 s. These biasing conditions ensure the injection of channel electrons into the various trap levels in the GaN buffer.[11,13] Consecutively, after the 1 s filling pulse, the I_D transient signals were measured as a function of time by ramping down the drain voltage (V_D) to 1.5 V while maintaining the gate voltage (V_G) at 0 V. To avoid self-heating effects, we chose a low V_D (1.5 V) for measurements. The schematic time domain representation of the applied $V_{D[ON]\_Stress}$ of 10 and 60 V and the corresponding I_D transients measured at (V_G, V_D) = (0, 1.5) V are shown in Figs. 1(c) and 1(d), respectively. Since the $V_{D[ON]\_Stress}$ facilitates current collapse due to buffer-related trapping, the I_D transients obtained after a $V_{D[ON]\_Stress}$ will be dominated



Trap E_a (eV)	Physical Origin	Ref
0.44	C/O/H Impurities	[23]
0.51, 0.55	Nitrogen Anti-sites	[24] [25] [26]
0.58, 0.65	VGa+Oxygen complex	[24] [27]
0.73	Nitrogen Interstitials	[28]

Fig. 2. Differential signals from the temperature-dependent I_D transients at (a) (0, 10), (b) (0, 40), and (c) (0, 60) V, and (d) the respective Arrhenius plots for determining the activation energies for HEMTs obtained under different $V_{D[ON]\_Stress}$ conditions (inset: trap E_a physical origin). Device dimensions: $L_{sg}/L_g/L_{gd}/W_g$ = 2/2/6/80 μm.

by the emission of electrons from various trap energy levels present in the heterostructure band gap.[7,8] The surface-related trapping effects[15] have been neglected since the surface traps were passivated by [$(NH_4)_2S_x$] surface treatment[2,10,21] followed by Si_3N_4 passivation.

The activation energies (E_a's) for the electron emission from the trap levels were determined by the Arrhenius analysis of the temperature-dependent emission time constants. On the basis of the measured I_D transients after the $V_{D[ON]\_Stress}$ of 10 to 60 V, with steps of 10 V, six different traps with E_a's of 0.44, 0.51, 0.55, 0.58, 0.65, and 0.73 eV, and corresponding capture cross sections of 4×10^{-21}, 2.6×10^{-20}, 8.5×10^{-20}, 3.5×10^{-19}, 2.4×10^{-18}, and 4.2×10^{-17} cm^2, respectively, were obtained. Figure 2 shows the temperature-dependent I_D transient signals under the following $V_{D[ON]\_Stress}$ conditions: (a) (0, 10), (b) (0, 40), and (c) (0, 60) V. Figure 2(d) shows the compiled Arrhenius plots showing the different E_a's obtained under different $V_{D[ON]\_Stress}$ conditions, and the physical origins[23–28] of the traps with their corresponding E_a's are shown in the inset. The obtained E_a's exhibited a linear dependence on the applied $V_{D[ON]\_Stress}$ conditions. This behavior suggests that the obtained E_a's may be from DTE levels rather than from discrete single trap energy levels. For a better understanding of the above behavior, a thorough analysis of the differential signals obtained from the I_D transients was performed as a function of measurement temperature and applied $V_{D[ON]\_Stress}$. Figure 3 shows the differential signals under all $V_{D[ON]\_Stress}$ conditions measured at (a) 60 and (b) 120 °C. For 60 °C, the signal peaks shifted towards higher time constants with increasing $V_{D[ON]\_Stress}$. However, the signal peaks shifted towards lower time constants for 120 °C under the $V_{D[ON]\_Stress}$ conditions of (0, 50) and (0, 60) V. Figure 3(c) shows the obtained emission time constant (τ_e) as a function of $V_{D[ON]\_Stress}$ at different temperatures. Under a lower $V_{D[ON]\_Stress}$ condition of (0, 40) V, τ_e increased linearly at all measurement temperatures. At measurement temperatures ≤80 °C, τ_e tends to saturate under $V_{D[ON]\_Stress}$ conditions of >(0, 40) V. However, τ_e started decreasing at higher temperatures (100 and 120 °C). This anomalous behavior under the combination of high $V_{D[ON]\_Stress}$ and high measurement temperature is attributed to self-heating effects in the AlGaN/GaN

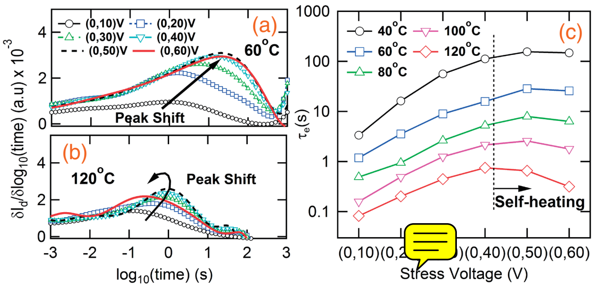


Fig. 3. Differential signals of the temperature-dependent I_D transients obtained under all $V_{D[ON]\_Stress}$ conditions at (a) 60 and (b) 120 °C, and (c) emission time constants vs $V_{D[ON]\_Stress}$ conditions for all measurement temperatures.

HEMTs. The effect of self-heating on τ_e due to bias stress and temperature was also reported by Albahrani et al.[29]

In order to understand the physics governing the trapping and detrapping mechanisms from DTE levels, 2D numerical simulations using TCAD were performed with the trap information obtained from the experiments. Poisson's equation along with the carrier continuity equations for electrons and holes was solved by taking into consideration the contribution of mobile and fixed charges and ionized traps. The dynamic traps are modelled by a Shockley–Read–Hall recombination term included in the continuity equations. For transient trap simulations, an additional differential rate equation is solved to account for emission and capture processes. As a filling pulse, both the source and gate terminals were biased at 0 V, and a drain bias ($V_{D[ON]\_Stress}$) of 10 to 60 V (with steps of 10 V) was applied for a period of 1 s, with a rise/fall time of 10 ns. For obtaining the I_D transient signal, the drain bias was ramped down from the $V_{D[ON]\_Stress}$ to a measurement voltage of 1.5 V while maintaining the source and gate terminals at 0 V. All the measurements were performed under an isothermal condition of 313 K (40 °C). No thermal effects were considered in the simulation. Thus, the model does not account for self-heating effects. For a discrete single quantum energy level, the total trap charge density (Q_{Tr}) in the single trap energy (STE) model is given by

$$Q_{Tr} = qN_{t,A} = q.n_t.F_{t,A}, \qquad (1)$$

where q, $N_{t,A}$, n_t, and $F_{t,A}$ are the electronic charge, ionized acceptor density, trap density, and probability of ionization, respectively. $N_{t,A}$ increases with increasing $V_{D[ON]\_Stress}$ due to the increasing spread of the 2DEG wave function beyond the 2DEG channel and, consequently, this increases $F_{t,A}$ [see Fig. 1(b)]. However, the dependence of τ_e on the applied $V_{D[ON]\_Stress}$ could not be validated through the STE model. Instead, a multitrap energy (MTE) model, consisting of several discrete energy levels in the heterostructure band gap, was incorporated into the TCAD environment. In the MTE model, the effective ionized acceptor density ($N_{t,A,eff}$) is given by

$$N_{t,A,eff} = \sum_{j=1}^{m} N_{t,A,j}, \qquad (2)$$

where m is the total number of acceptor-like trap levels present in the energy band gap and $N_{t,A,j}$ is the ionized acceptor trap density for the jth trap energy level. The probability of ionization ($F_{t,A,j}$) and the electron emission rate ($e_{n,A,j}$) for the jth trap energy level, $E_{t,A,j}$, are given by

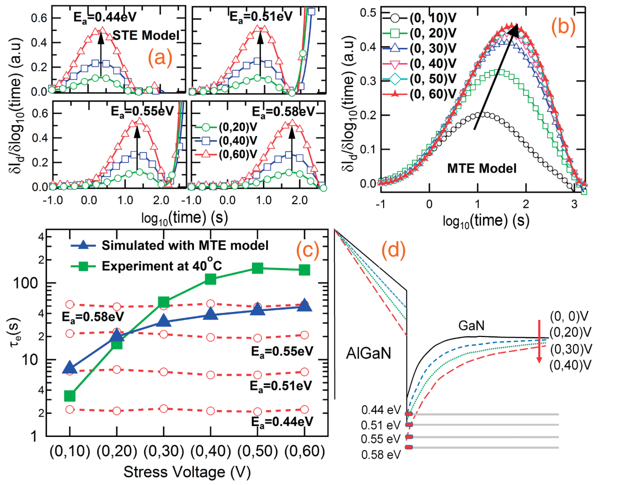


Fig. 4. Differential signals of the I_D transients obtained from the TCAD simulations (a) for individual trap levels under different $V_{D[ON]\_Stress}$ conditions; (b) for multitrap levels; (c) emission time constants vs $V_{D[ON]\_Stress}$ conditions for all measurement temperatures for individual traps using STE model (broken lines), MTE model (filled triangles), and compared with the experimentally obtained emission time constant at 40 °C (filled squares); (d) representation of AlGaN/GaN heterojunction energy band bending under different $V_{D[ON]\_Stress}$ conditions and the distribution of traps in the GaN energy band gap.

$$F_{t,A,j} = \frac{v_n \sigma_{c,n,j} n + e_{p,A,j}}{v_n \sigma_{c,n,j} n + v_p \sigma_{c,p,j} p + e_{n,A,j} + e_{p,A,j}}, \qquad (3)$$

$$e_{n,A,j} = g.v_n \sigma_{c,n,j} n_i \exp\left(\frac{E_{t,A,j} - E_i}{kT}\right), \qquad (4)$$

where $\sigma_{c,n,j}$ and $\sigma_{c,p,j}$ are the capture cross sections and v_n and v_p are the thermal velocities for electrons and holes, respectively, and g is the degeneracy factor of the trap center.[20] For validating the experimentally observed $V_{D[ON]\_Stress}$ dependence of E_a, 2D transient simulation was performed by separately considering the STE and MTE models. Figure 4 shows the differential signals of the I_D transients obtained from the TCAD simulations with (a) the STE model with individual trap levels and (b) the MTE model, under different $V_{D[ON]\_Stress}$ conditions. No considerable peak shift is observed for the differential signals when simulated with the STE model. However, the signal obtained with all the traps incorporated together (MTE model) shows only one significant peak that shifted towards a higher time constant with increasing $V_{D[ON]\_Stress}$, as also observed experimentally [see Fig. 3(a)]. For a better understanding, a comparison of the τ_e vs $V_{D[ON]\_Stress}$ characteristics between the experiment and the simulation is shown in Fig. 4(c). When simulated with individual trap levels using the STE model, the corresponding τ_e (broken lines) shows no variations with increasing $V_{D[ON]\_Stress}$. This is because the emission rate $e_{n,A,j}$ does not change significantly with $V_{D[ON]\_Stress}$, as given by Eq. (4). However, the transient simulations with all traps incorporated together (MTE model) resulted in an effective τ_e (filled triangles), which increased with increasing $V_{D[ON]\_Stress}$. This effective τ_e ranged between the τ_e values of the upper and lower limits of the incorporated discrete trap energy levels. As shown in Fig. 4(c), the rate of increase in τ_e with respect to $V_{D[ON]\_Stress}$ exhibited a similar trend for both experiment (filled squares) and simulation (filled triangles), except with some deviation at high $V_{D[ON]\_Stress}$ caused by self-heating

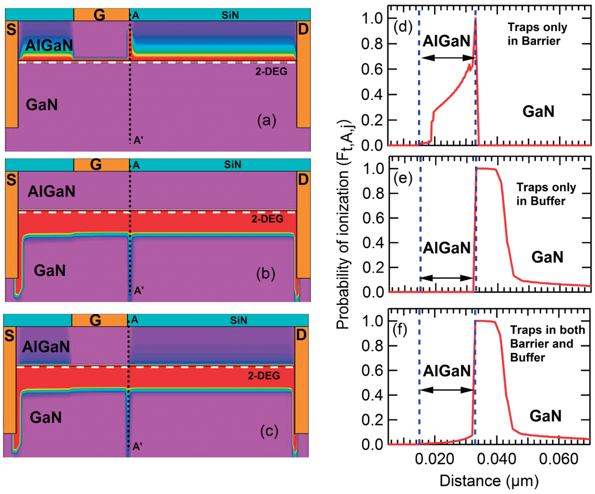


Fig. 5. TCAD simulated probability of ionization ($F_{t,A,j}$) with the traps incorporated in only the (a) AlGaN barrier or (b) GaN buffer, and (c) in both AlGaN barrier and GaN buffer; (d)–(f) show the corresponding $F_{t,A,j}$ at the cut-line position of AA′, under the applied $V_{D[ON]\_Stress}$ of (0, 60) V.

effects.[29] This shows that the obtained transient signals contain information from various trap energy levels that are activated by the applied $V_{D[ON]\_Stress}$.[30]

Thus, it is clearly evident from both experiments and simulations (MTE model) that τ_e increases with $V_{D[ON]\_Stress}$. This effect can be explained only if we consider the existence of a DTE function rather than discrete quantum energy levels in the GaN energy band gap, as shown in Fig. 4(d). As $V_{D[ON]\_Stress}$ increases, the GaN conduction band bends downwards, activating different trap energy levels.[30] The TCAD simulated probability of ionization ($F_{t,A,j}$) under the applied $V_{D[ON]\_Stress}$ of (0, 60) V is shown in Fig. 5 with the traps incorporated in only the (a) AlGaN barrier or (b) GaN buffer, and (c) in both the AlGaN barrier and the GaN buffer. Figures 5(d)–5(f) show the corresponding $F_{t,A,j}$ at the cut-line position of AA′. It is evident that when traps are incorporated in both the barrier and the buffer, the probability of electron trapping under ON-state stress conditions is higher in the GaN buffer.[9,11,12,15] This has also been confirmed by the hydrodynamic model.[11] In the STE model, $e_{n,A}$ would not change with varying $V_{D[ON]\_Stress}$. Considering the MTE model, $N_{t,A,eff}$ increases with increasing $V_{D[ON]\_Stress}$ and can be expressed as

$$N_{t,A,eff} = n_t \sum_{j=1}^{m} F_{t,A,j} \cong n_t \cdot \frac{v_n \sigma_{c,n,eff} n}{v_n \sigma_{c,n,eff} n + e_{n,A,eff}(V_{dq})}, \quad (5)$$

where $e_{n,A,eff}(V_{dq})$ is the new effective emission rate corresponding to the effective activation energy (E_{a_eff}) that is a function of the applied $V_{D[ON]\_Stress}$. Thus, the increase in τ_e with the increase in $V_{D[ON]\_Stress}$ is due to the presence of DTE levels in the GaN energy band gap. An accurate functional dependence of emission rate or E_a on the applied stress voltage requires further investigation.

Temperature-dependent current-transient-based trap analysis under fully ON-state stress conditions has been carried out for AlGaN/GaN HEMTs on Si. Six different trap activation energies (0.44, 0.51, 0.54, 0.58, 0.65, and 0.73 eV) were obtained under six different $V_{D[ON]\_Stress}$ conditions with varying capture cross section. The dependence of τ_e on $V_{D[ON]\_Stress}$ was observed in both experiments and numerical simulations. A clear distinction between the STE and MTE models has also been presented through this study. The dependence of the obtained τ_e on the applied $V_{D[ON]\_Stress}$ could be simulated only by incorporating the MTE model. This clearly indicates the presence of continuous DTE levels in the GaN energy band gap. Such a quantitative approach towards trap characterization will help device designers in developing a more accurate reliability model for GaN-based HEMTs.

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